

B. Tech Degree III Semester Examination November 2010**ME/EB/EI/IT/EC 302 ELECTRICAL TECHNOLOGY**
(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART - A
(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Explain briefly the operation of a transformer and sketch phasor diagram on no – load.
 (b) Derive the expression for saving of copper in an auto transformer as compared to an equivalent two winding transformer.
 (c) Enumerate the various losses in a d.c machine. Derive the condition for maximum efficiency of a d.c generator.
 (d) Draw and explain different characteristics of a d.c series motor.
 (e) Define voltage regulation of an alternator. Explain the various factors which may affect the regulation of an alternator.
 (f) Discuss why single phase induction motors do not have a starting torque.
 (g) Give an account of different electrical equipments in power stations.
 (h) What are the advantages of high voltage transmission? State its limitations also.

PART – B

(4 x 15 = 60)

- II. Obtain the equivalent circuit of 1ϕ , 4 KVA, 200/400 V, 50 Hz transformer from the following test results as referred to ℓ_v side.
 OC test : 200 V 0.7 A 70 W on ℓ_v
 SC test : 15 V 10 A 80 W on hv
 Also calculate the efficiency of the transformer at half load, 0.8 pf. (15)
- OR
- III. (a) Draw a complete phasor diagram for a practical transformer when the load is inductive. (7)
 (b) What are instrument transformers? Where are they used? (8)
- IV. A 500 V d.c shunt motor has armature and field resistances of $1.2\ \Omega$ and $500\ \Omega$ respectively. When running on no – load, the current taken is 4 A and the speed is 1000 rpm.
 (i) Calculate the speed when the motor is fully loaded and the total current drawn from the supply is 26 A.
 (ii) Estimate the speed at this load if
 (1) a resistance of $2.3\ \Omega$ is connected in series with the armature.
 (2) the shunt field current is reduced by 15%. (15)
- OR
- V. (a) What is armature reaction? What are the effects of the armature reaction? (7)
 (b) Explain the methods of speed control of d.c shunt motors. (8)
- VI. A 100 KVA, 3000 V, 50 Hz, 3ϕ star connected alternator has effective armature resistance of $0.2\ \Omega$. The field current of 40 A produces short circuit current of 200 A and an open circuit emf of 1040 V (line value). Calculate the full load voltage regulation at (i) 0.8 pf lagging (ii) 0.8 pf leading. (15)
- OR
- VII. (a) Explain the working principle of a 3 phase induction motor. Why does an induction motor never run on synchronous speed? (6)
 (b) Derive the torque – slip characteristics of a 3 phase induction motor. (6)
 (c) What is a synchronous condenser? (3)
- VIII. With necessary schematic diagram explain the working of a hydro electric power plant. (15)
- OR
- IX. Give a schematic arrangement and salient features of a nuclear power station. (15)